Raúl Beltrán Gómez

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Professional Summary

AI and software engineer with production experience in computer vision (YOLO detection and pose models, ONNX and TensorRT deployment, real-time industrial vision) and in LLM systems (MCP servers, agents, NLP, REST APIs). Strong background in Python backends, secure integrations and cross-stack development (React, Unity), with hands-on integration of industrial robots and PLCs.

Experience

Promart Automatizaciones — Madrid, Spain
Computer Vision and AI Engineer — *01/06/2026 – Present*

- Built the real-time vision system of a robotic teat-sealing cell (rotary milking parlour): YOLOv8 teat detection (XYZ mm), YOLOv8-Pose posture safety gate, OCR, cow ID; ONNX/TensorRT GPU inference.
- Owned the model lifecycle (labeling, augmentation, Optuna tuning, ONNX/TensorRT export, mAP and latency benchmarks, S3 versioning, CI/CD) and its Python stack: clean architecture, FastAPI, tests.
- Wrote a Python robot simulator running the production program against the PLC; also integrated the cell: conveyor-synced Yaskawa YRC1000 (INFORM) and Omron NX1P2 PLC (EtherNet/IP, FINS).

Cogitia — Madrid, Spain (Remote for Barcelona, Spain)
MCP, LLM Agents, Chatbot and REST API Developer — *14/05/2025 – 20/05/2026*

- Designed and maintained MCP servers for enterprise workflows using embedded LLM clients and robust prompt, tool and resource patterns.
- Developed intelligent LLM agents with LangGraph-style orchestration for business automation workflows.
- Built NLP modules for automation, including data operations, scheduling, and internal query handling.

Word Games — Havana, Cuba (Remote for Bogotá, Colombia)
Game Developer, Development Team — *20/10/2024 – 10/05/2025*

- Led end-to-end programming of Arcane Ascent (deck-building roguelike), shipped on Steam Early Access.
- Integrated native Steam systems (achievements, saves), improving platform readiness and user experience.

Devman — Havana, Cuba (Remote for Madrid, Spain)
NLP, Chatbot and REST API Developer — *10/04/2024 – 14/10/2024*

- Designed an e-commerce assistant with NLP intent handling and product retrieval for faster discovery.
- Built backend services in FastAPI, including API key security, WebSocket chat, and authorization layers.

Education

University of Havana — Havana, Cuba
Bachelor of Computer Science — Expected 23/01/2024

- Relevant fields: Artificial Intelligence and Machine Learning, Information Retrieval, Compilers, Operating Systems, Databases, Distributed Systems, Networks, Algorithms and Data Structures, Graph Theory.

Skills

Languages: Spanish (native), English (B1).
Programming: Python, C#, C/C++, JavaScript/TypeScript, Assembly (MIPS).
Technical: Computer Vision (YOLO, OpenCV, ONNX Runtime, TensorRT, OCR), PyTorch, TensorFlow, Optuna, MCP, LLMs, LangGraph, LangChain, NLP, RAG, FastAPI, .NET, Databases, ReactJS, PyQt6, Unity Engine, Docker.